Acquisition Is Not Closure: Fail-Closed Control for Open-World Scientific-Literature Discovery

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Abstract

Scientific-literature discovery combines evidence acquisition with decisions about when a search may stop. These functions are often evaluated as one policy, although a locally exhausted or unavailable retrieval route does not establish that the scientific task is complete. We introduce a fail-closed control layer that separates route stopping from task closure, derives route independence from backend and query provenance, distinguishes content identity from question-conditioned processing state, and retains unresolved material routes as open obligations. A 390-task complete-gold controlled index shows the intended mechanism behavior: the governed controller reaches mean recall 0.979487 with no premature task closure, but the preregistered comparison is underpowered and does not establish superiority. An exact-contract battery isolates the unresolved-route rule on 400 protected cases. The fail-closed controller is correct on 400 cases, compared with 250 for a donor-complete product that excludes unavailable material routes; an information-equivalent product ties the controller. The registered external Text REtrieval Conference COVID-19 (TREC-COVID) gate fails. Against a BM25 lexical baseline, recall@100 is lower by 0.01769 with a 95% bootstrap interval of $[-0.02729, -0.00906]$, and the controller reads 175.7% more candidates. Although normalized discounted cumulative gain at rank 10 (nDCG@10) is higher by 0.1488, that secondary result cannot replace the registered recall-and-cost criteria. The contribution is therefore a bounded control method for representing acquisition authority, not evidence of external retrieval superiority or open-world completeness.

Keywords: scientific literature discovery; information retrieval; systematic review; stopping rules; coverage; open-world search; fail-closed control.

1 Introduction

Scientific-literature discovery systems increasingly combine lexical and semantic retrieval, query reformulation, citation expansion, selective reading, screening, and learned stopping. Better acquisition can recover more useful material, but it cannot answer a different question by itself: what does the observed search state authorize a system to claim about completeness? A route may be locally exhausted while another route remains untried. A provider may be unavailable rather than empty. Two differently named routes may share the same backend and query derivation. A document may have been encountered before but not processed for the current scientific question. Treating these cases as ordinary negative evidence creates false closure.

This paper studies that control boundary rather than proposing a new general retriever. The method separates an acquisition policy, which selects the next search or reading action, from a closure authority, which decides whether the current evidence state permits route stopping or task closure. Four design elements implement the separation. First, route independence is supported by backend, query-derivation, and capture provenance rather than route labels. Second, route-local stopping never certifies task-global completeness. Third, unavailable, censored, or provider-invalid material routes remain explicit open obligations. Fourth, content identity is recorded separately from question-conditioned read state, preventing duplicate discovery from inflating coverage without suppressing legitimate rereading.

The empirical question is deliberately narrow: do these control elements behave in their intended directions under complete denominators and protected exact contracts, and do they support an external superiority claim? We use a complete-gold controlled index to expose premature closure and mechanism failures, then an exact-contract battery to isolate unresolved-route authority. The external decision is supplied by a preregistered TREC-COVID comparison. Its recall and cost gates both fail. We retain the favorable nDCG result as a secondary observation, but do not use it to rescue the registered decision.

The resulting contribution is a methods and critical system-design claim. It does not claim that finite open-world search proves completeness, that the controller outperforms established retrieval systems, or that existing retrieval, screening, stopping, and memory components are novel. Instead, it provides a composable representation of what acquisition outcomes are allowed to certify, together with controlled evidence and an adverse external test that fix the present boundary.

2 A separate authority layer

Let x_t denote the observable search state at step t, and let $a_t \sim A(x_t)$ be an action selected by an acquisition policy A. The action may issue a query, expand a citation neighborhood, fetch a document, reread a source under a new extraction frame, or stop a route. Acquired material updates an evidence state E_t and a typed obligation state O_t. A separate closure map

$$C(E_t, O_t) \in \{\textsc{continue}, \textsc{route-stop}, \textsc{open}, \textsc{undetermined}, \textsc{task-close}\}$$

decides what the combined state is allowed to certify.

The separation is intentionally asymmetric. Acquisition is permissive: a stronger lexical, dense, reasoning-aware, field-aware, or graph-expansion method can replace a weaker method. Closure is conservative: a utility score, coverage estimate, confidence value, sufficiency judgment, or provider failure cannot delete an unresolved material obligation. This interface makes retrieval and stopping advances composable without granting them authority they were not evaluated to carry.

Four state distinctions are load-bearing. Route identity separates backend, query derivation, and capture provenance. Work identity joins equivalent source identifiers, whereas content identity distinguishes materially different renditions. Read state records the question and extraction schema under which a content item was processed. Route state distinguishes exhaustion from transport failure and task closure from every route-local terminal. Section 3 defines these objects and the failure direction that each distinction prevents.

The method is fail-closed, not pessimistic by definition. A task can close when every declared material obligation is discharged. An unavailable nonmaterial route does not block closure. A route becomes relevant to the decision only when the frozen task contract marks its unresolved evidence as material. This materiality condition is tested explicitly in the protected exact-contract battery rather than inferred after outcomes.

3 Formal control model

3.1 Work, content, and read identity

Let R be the raw identifiers emitted by retrieval routes. Canonicalization maps each identifier to a scheme-qualified work key and a rendition version. Alias-connected keys form a work. Each retrieved rendition also carries a content digest c. Deduplication and overlap are computed over content digests, not locators, because reminted identifiers must not make two captures appear independent.

A read receipt is a tuple (w, c, s, f) containing work, content digest, extraction-schema version, and scientific frame. A candidate is already read only when all four coordinates match an existing receipt. A changed content digest, schema, or frame creates a legitimate reread. This rule separates duplicate processing from processing that is newly required by the scientific question.

3.2 Earned route independence

Definition 1 (Route capture)**.** A route capture is $\rho = (k, b, q, C)$, where k is route kind, b is backend identity, q is the query-derivation process, and C is the captured set of content identities.

For two route captures, the controller accepts an independence claim only if route kind, backend identity, and query derivation differ:

$$\mathrm{ind}(\rho_i, \rho_j) \iff k_i \neq k_j \wedge b_i \neq b_j \wedge q_i \neq q_j.$$

This predicate is a conservative design requirement, not a proof of statistical independence. Its purpose is to reject obvious shared-generation causes before overlap is interpreted. Zero observed overlap alone does not establish independence.

For an accepted pair with capture sizes n_i, n_j and overlap m_{ij}, a two-source diagnostic may report $\widehat{N} = n_i n_j / m_{ij}$. The diagnostic is undefined when $m_{ij} = 0$. In every case it remains advisory because adaptive search violates the interpretation of independent capture occasions unless a separate design establishes it.

3.3 Route control and task closure

Each route records attempts, retrieved and novel content identities, resource use, failure signatures, and backend exhaustion. A priority-ordered policy may stop on budget, stop on reported exhaustion, suspend on transport failure, switch after a flat streak, reformulate, or continue. Backend exhaustion is a claim about the index. Transport failure is a claim about the channel. The latter suspends the route and preserves an open obligation.

Proposition 1 (Route-stop non-certification)**.** *No route decision certifies task completeness. Task closure is a separate decision over the ensemble of declared material obligations.*

The proposition is an implementation invariant rather than an empirical law. Its consequence is testable: if a route-stop signal is promoted directly to task closure, the evaluator should observe premature closure whenever another route still contains reachable relevant material. Task closure is permitted only when all declared material routes are discharged or explicitly resolved by the task contract, no transport failure leaves a material obligation open, and no advisory coverage diagnostic supplies the authority.

3.4 Observable error families

The model defines four evaluator-visible failures. A route-stop false positive occurs when a stopped route can still reach relevant material. A route-stop false negative occurs when search continues beyond route exhaustion. Premature task closure occurs when relevant material remains reachable within the task contract. Duplicate processing occurs when all four read coordinates repeat; changed content, schema, or frame instead defines a legitimate reread. Complete gold is required wherever these definitions depend on the full relevant set.

4 Methods

4.1 Evidence boundary and prospective discipline

The evaluation separates three evidence classes. The complete-gold controlled index tests mechanism behavior under an enumerable denominator. The protected exact-contract battery isolates closure decisions when route state and materiality are known. The TREC-COVID experiment tests the registered external recall-and-cost claim on a public collection. Results from one class cannot promote another: synthetic mechanism behavior does not establish external retrieval superiority, and a secondary TREC-COVID metric cannot replace the registered primary criteria.

For each result-bearing campaign, task identities, controller and comparator identities, budgets, evaluator rules, decision thresholds, and repeat or bootstrap seeds were frozen before outcomes were inspected. Candidate systems received only public task views and label-free retrieved records. Complete gold was available only to the evaluator. Post-outcome analysis could describe failure stages or generate manuscript tables, but could not change the decision rule.

The controller returns one of three scientific outcomes. A task passes when all declared material obligations are discharged. It fails when the system makes a checkable incorrect decision or exhausts the registered budget without a valid closure. It remains undetermined when material evidence cannot be assessed, such as when a required route is unavailable. An undetermined outcome is not a zero and is not evidence of absence.

4.2 Complete-gold controlled index

The controlled index contains 390 tasks derived from 78 topics and 1,210 synthetic documents. Relevant items are distributed across lexical, semantic, citation, reformulation, and restricted-provider routes. The generator includes shared backends, duplicate locators, content revisions, extraction-question shifts, and unavailable routes. Budgets are set below exhaustive probing so route allocation and stopping remain observable.

Fourteen systems were frozen: the governed controller, six confirmatory baselines, one exploratory adaptive comparator, and six ablations. Confirmatory baselines include single-route lexical and dense retrieval, one-pass hybrid retrieval, a protocol-driven systematic-review policy, and stopping controls. All systems use the same host-owned session, route interface, resource meter, and evaluator. Three deterministic repeat seeds test execution stability. They do not increase the statistical unit beyond the 390 tasks.

The main descriptive measures are complete-gold recall, premature task closure, pass rate, undetermined rate, budget-exhaustion rate, duplicate processing, legitimate rereading, route contribution, and route overlap. Route-stop false positives and false negatives are evaluated separately from task closure. The frozen precision plan requires 385 tasks for its committed tier; the assembled 390 tasks meet that count. However, the achieved half-width of 0.0496 exceeds the frozen superiority margin of 0.03. The primary controlled comparison is therefore labeled underpowered and cannot authorize a superiority claim.

4.3 Exact-contract battery

The protected battery contains 400 cases across five acquisition settings: scientific literature, web or API evidence, dataset registries, code repositories, and experiment or measurement routes. Cases distinguish complete obligation discharge, one available route remaining open, material routes that are unavailable, censored, or provider-invalid, duplicate coverage, question-processing obligations, and low-utility routes that remain material. Held-out variants add unavailable nonmaterial decoys, preventing a blanket rule that any unavailable source blocks closure.

The primary comparator receives strong donor mechanisms: evidence coverage, decision sufficiency, utility stopping, finite route certificates, deduplication, route state, materiality, and question-conditioned processing. It makes the same route-local decisions as the fail-closed controller. The only systematic difference is task aggregation. After all available material routes stop, the comparator excludes unavailable, censored, or provider-invalid material routes from the closure denominator. A second comparator permits a global sufficiency signal to close the task. A third comparator receives the same unresolved-route semantics as the controller and defines an information-equivalent boundary expected to tie.

The primary effect is the paired difference in exact scientific-closure decisions between the fail-closed controller and the donor-complete available- route product. The frozen practical margin is 0.10. Uncertainty uses a domain-stratified bootstrap with 20,000 resamples, and discordance is reported with exact McNemar counts. Nonregression requires zero false task closures, perfect closure on clean cases, no unnecessary undetermined decisions, and no materiality error on held-out decoys.

4.4 Registered TREC-COVID comparison

The external comparison uses the 50 TREC-COVID topics and the public test collection [22]. Four arms were frozen: BM25, a reciprocal-rank-fusion hybrid, the governed controller, and a diagnostic variant that ignores unresolved route obligations when declaring completion. Each arm is evaluated on the same topic set. The topic is the paired inference unit.

The registered pass rule is noncompensatory. First, recall@100 must be noninferior to the best comparator with margin -0.02, assessed using the lower bound of a 95% paired bootstrap interval from 10,000 resamples. Second, the controller must reduce candidates or tool calls by at least 25%, with the confidence interval's lower bound showing at least a 10% reduction. Third, premature-stop error must not increase. Failure of either recall or cost is sufficient to fail the gate.

nDCG@10 was measured but was explicitly outside this pass rule. It describes the ranking quality of the first ten results and can therefore move differently from recall@100 and reading cost. We report it as secondary evidence only. The registered gate is not amended after observing the favorable nDCG result.

4.5 Bounded external stress tests

Additional external probes are used for stage attribution and adverse-case retention, not for the headline decision. A MetaSyn probe runs pinned BM25 retrieval and a deterministic public-protocol screening rule on all 86 released test reviews, then applies the official identifier-only evaluator. A bounded AutoResearchBench Deep probe uses the released title judge with a separate nine-case positive and negative control. OpenAIRE/Crossref campaigns enforce a provider-validity threshold before any comparison can be interpreted. Public screening studies retain frozen active-learning donors and prospectively fixed primary recall and work-saving criteria. Each probe keeps its own denominator, authority, and failure state.

System	Recall	Premature closure	Pass	Undetermined	Fail
Governed controller	0.979487	0.00	0.67	0.18	0.15
Protocol-driven review	0.666667	1.00	0.00	0.00	1.00
Sparse–dense hybrid	0.555556	1.00	0.00	0.00	1.00
One-pass retrieval	0.555556	1.00	0.00	0.00	1.00
Lexical retrieval	0.333333	1.00	0.00	0.00	1.00
Dense-route comparator	0.333333	1.00	0.00	0.00	1.00
Agentic single route	0.333333	1.00	0.00	0.00	1.00

Table 1: Descriptive complete-gold results over 390 tasks. Failure for the governed controller is honest budget exhaustion, not premature closure. The primary comparison is underpowered under the frozen plan and is not interpreted as superiority.

5 Results

5.1 Controlled-index behavior

The governed controller reached mean complete-gold recall 0.979487 and made no premature task closures. It closed 260 of the 390 tasks correctly. It failed 59 tasks after exhausting the frozen budget without claiming completeness, and it left 71 tasks undetermined because an unavailable route still held relevant material. The strongest confirmatory comparator was the protocol-driven systematic-review policy, with mean recall 0.666667. That comparator closed all 390 tasks while relevant material remained reachable. The descriptive recall difference was 0.312821, but the frozen plan's underpowered label prevents this comparison from supporting superiority.

The governed and protocol-driven controllers were identical through the first three search calls. By query 7, mean recall was 0.757835 for the governed controller and 0.666667 for the comparator. The comparator remained at 0.666667 through query 12, whereas the governed controller reached 0.979487. The descriptive difference therefore arose after the shared early search prefix, when open obligations led to additional acquisition.

5.2 Mechanism ablations

The safety ablations failed in their preregistered directions. Removing the route-independence check reduced mean recall to 0.444444 and caused premature closure on every task. Allowing a route stop to certify task closure produced mean recall 0.333333 and the same 100% premature-closure rate. Letting an advisory coverage diagnostic control closure produced mean recall 0.555556 and again closed every task prematurely. These ablations test the observability of forbidden transitions; they are not competitive retrieval baselines.

Removing content-identity deduplication increased the duplicate-processing rate from 0 to 0.31453. It caused 193 budget-exhaustion failures and reduced pass rate from 0.666667 to 0.369231. Removing the question coordinate from the read ledger left aggregate recall unchanged at 0.979487, but reduced legitimate rereads in the extraction-shift cases from 3.666667 to 1.0. This is a mechanism effect, not a recall gain. Removing the unavailable-route open state also left recall unchanged, but converted all 71 censored cases from undetermined outcomes into asserted completions and premature-closure failures.

The separate route-stop replay recorded 13 false-positive route stops among 1,950 route-stop events (0.006667). These local errors did not become task-level false closures because the unresolved route obligation remained open. Among 1,878 task-route instances that reached the evaluator's

Controller	Correct / 400	False task closures	Clean closure
Fail-closed control	400	0	1.000
Available-route donor product	250	150	1.000
Global-sufficiency product	100	300	1.000
Information-equivalent product	400	0	1.000

Table 2: Protected exact-contract battery. The information-equivalent product receives the same unresolved-route semantics and ties the fail-closed controller. The result therefore isolates an authority rule rather than inherent model expressivity or centralization.

Arm	recall@100	nDCG@10	Mean reads	Mean route calls
BM25	0.110334	0.417992	85.52	1.0
Reciprocal-rank fusion	0.077365	0.489570	170.30	2.0
Governed controller	0.092642	0.566746	235.80	3.0
Diagnostic closure variant	0.092642	0.566746	235.80	3.0

Table 3: External TREC-COVID outcomes over 50 paired topics. The registered decision used recall noninferiority and cost reduction. nDCG@10 was measured but was not a gate criterion.

exhaustion point, no route-stop false negative was observed. This null result is retained and does not establish a generally optimal stopping rule.

5.3 Exact-contract isolation of unresolved-route authority

The fail-closed controller produced the correct scientific-closure decision in all 400 protected cases. The donor-complete available-route comparator was correct on 250 cases, and the global-sufficiency comparator was correct on 100. The paired difference against the donor-complete comparator was 0.375, with a 95% domain-stratified bootstrap interval of $[0.3275, 0.4225]$. Exact McNemar discordance was $b = 150, c = 0$. All 150 comparator errors were the material unavailable, censored, or provider-invalid cases that its aggregation rule removed from the denominator.

The nonregression conditions also passed. The fail-closed controller made no false task closure, closed every clean case correctly, produced no unnecessary undetermined outcome on clean cases, and made no materiality error on 80 held-out nonmaterial-decoy cases. Performance was identical across the five acquisition settings. Because the information-equivalent product also achieved 400 correct decisions, the evidence supports the unresolved-route rule only. It does not support ownership of generic stopping, obligation, or closure machinery.

5.4 The registered TREC-COVID gate fails

BM25 was the best comparator for the registered recall-and-cost decision. Its mean recall@100 was 0.110334, compared with 0.092642 for the governed controller. The paired mean difference was -0.01769. The point estimate lay inside the -0.02 noninferiority margin, but the 95% bootstrap interval $[-0.02729, -0.00906]$ did not. Noninferiority was defined by the interval, so the recall criterion failed.

The cost criterion also failed. The governed controller read 235.8 candidates per topic on average, compared with 85.52 for BM25, an increase of 175.7% rather than the required 25% reduction. Its mean route calls were 3.0 rather than 1.0. The overall registered gate therefore failed on both recall and cost.

Evidence family	Observed result	Allowed interpretation
MetaSyn public probe	Retrieval recall 0.7485; inclusion recall 0.5651	Bounded stage attribution only
AutoResearchBench Deep	0/600 title matches after a 9/9 judge control	Bounded negative candidate-generation diagnostic
OpenAIRE/Crossref	Provider validity below the frozen 0.90 threshold	Comparison remains undetermined; zero differences are not equivalence
Public screening transport	Frozen donor outperforms the candidate on primary recall and work-saving endpoints in multiple development studies	Adverse within-pool evidence; not an acquisition or closure test

Table 4: External results retained at separate authority levels. No row is averaged with the TREC-COVID decision or promoted into system superiority.

nDCG@10 moved in the favorable direction. The governed controller exceeded BM25 by 0.1488, with a 95% bootstrap interval of $[0.1008, 0.1986]$, and won on 42 of 50 topics. This result indicates stronger top-ten ranking quality under the evaluated arm. It cannot compensate for the failed recall-and-cost gate, and no threshold or endpoint was changed after observing it.

5.5 External stress-test boundaries

The MetaSyn probe completed all 86 released test reviews. Macro retrieval recall was 0.7485 and final inclusion recall was 0.5651. Of 1,677 reference links, 598 were absent from the retrieved pools and another 343 were lost during screening, leaving 941 final false negatives. This result attributes loss within a bounded retrieval-and-screening probe; it does not evaluate the full controller.

The AutoResearchBench Deep probe returned no official title match across 600 tasks after its positive and negative judge control passed all nine cases. A separate stage analysis found negligible exact and substring recovery in the candidate sets, making candidate generation rather than the judge the bounded failure location. The result is not a matched baseline comparison.

The matched OpenAIRE/Crossref campaigns failed provider-validity thresholds. Their observed paired differences were zero because retrieval operated near a measurement floor or the route transport was invalid. Resampling identical near-zero differences yields a narrow interval regardless of scientific informativeness, so no equivalence or noninferiority conclusion is drawn. These undetermined outcomes and the adverse public screening results remain visible as limits on external generalization.

6 Related work

Scientific-discovery benchmarks show that both targeted and open-ended literature search remain difficult for current agents [25, 10]. Acquisition methods are already heterogeneous. Strong lexical retrieval can outperform large-model retrievers inside deep-research workflows [9]; reasoning traces can become retrieval signals [3]; fielded admission and selective section fetching can improve the accuracy–context trade-off [23]; and bibliography expansion can recover studies missed by API-only search [20]. These methods enlarge the acquisition layer. They do not, without an additional argument, establish task completeness.

Automated evidence synthesis provides a second neighboring literature. MetaSyn separates retrieval from eligibility screening, making stage-specific false negatives visible [24]. AgentSLR evaluates an end-to-end systematic-review workflow over expert-annotated review tasks [15]. Technology-assisted review has long studied active screening protocols, autonomy, recall targets, and workload reduction [5, 6, 21]. The present work adopts stage attribution and fair-budget comparison, while focusing on the authority transition between a local acquisition outcome and a global closure claim.

Stopping research includes statistical stopping for systematic review [1, 12, 27], decision-theoretic continuation [8, 26], verification-aware completion [18, 4], conformal coverage for adaptive reasoning [29], and structured sufficiency or gap judgments [7, 13]. Finite obligation certificates and evidence-bound workflow transitions offer still stronger donor mechanisms [2, 28]. We grant these mechanisms to the acquisition policy and ask a residual question: when a material route is unavailable, censored, or provider-invalid, may it disappear from the task-closure denominator? The exact-contract comparison is designed to change only that aggregation rule.

Capture–recapture methods and information foraging motivate, but do not settle, the same boundary. Capture–recapture estimates require defensible capture occasions and can be misleading under dependence [11, 19]. Information foraging describes the value of continuing or leaving a local information patch [16, 17]. In adaptive retrieval, however, a low-yield patch or a dependent route cannot be promoted into proof that the global literature has been exhausted. Question-conditioned memory likewise helps organize evidence [14], but the method here claims only the bookkeeping distinction between content identity and the question for which that content was processed.

7 Discussion and limitations

The controlled studies support a control-layer interpretation, not a retrieval leaderboard claim. In the complete-gold index, the governed controller maintains open obligations where the evaluator knows relevant material remains reachable. The ablations show that forbidden authority transitions are observable, while the exact-contract battery isolates the aggregation error made when unavailable material routes are excluded. These results demonstrate implementation behavior under authored contracts. They do not estimate real-web difficulty or prove that the declared route registry contains every material route.

The TREC-COVID result is the strongest external discriminator in the present paper because it tests the registered comparative claim on a public collection [22]. It is adverse. The recall noninferiority interval crosses the registered margin, and resource use moves in the wrong direction. The higher nDCG@10 is scientifically informative because it indicates better top-ranked relevance on many topics, but it answers a different question. A post-outcome substitution of nDCG for recall and cost would change the study rather than pass it. The paper therefore makes no external superiority claim.

Several additional probes remain bounded or undetermined. A public MetaSyn probe quantifies retrieval and screening loss, but it does not exercise the complete controller. An AutoResearchBench Deep probe returns no official title matches after a successful judge control, which is a bounded negative diagnostic rather than a matched comparison. OpenAIRE/Crossref campaigns fail their provider-validity criterion; their zero paired differences cannot be read as equivalence. Public screening studies sometimes favor the frozen donor over the candidate. Table 4 keeps these results visible without averaging incompatible authority levels.

The formal control rules also have limits. Route independence is conservative and provenance-based; it does not prove statistical independence. The closure decision is relative to declared material routes and reopen conditions. A previously unknown source family can therefore reopen a task. Content hashes support deduplication but do not decide scientific relevance. Finally, the controlled index supplies complete denominators by construction. Open-web completeness generally does not.

These limits shape the intended use. The control layer can sit beneath stronger retrievers, screening models, and stopping policies, because it constrains what their outputs mean rather than how they are produced. Its practical value is to prevent provider failure, local flatness, high estimated coverage, or apparent sufficiency from becoming an unsupported global claim. Whether this conservatism improves real scientific decisions at acceptable cost requires a new prospectively frozen external study.

8 Data, code, and reproducibility

A blinded review archive accompanies the manuscript. It contains the exact manuscript source, the synthetic complete-gold task specification, frozen protocols and summaries, protected-result receipts, TREC-COVID result records, figure sources, and the code needed to regenerate the controlled analysis. The archive includes a manifest with a cryptographic digest and byte count for every file. Third-party benchmark corpora are not redistributed; their public identifiers, pinned source revisions, and licence or access conditions are recorded in the manifest.

The controlled index is fully synthetic and regenerates from its recorded seed. A clean reproduction reconstructs all 16,380 normalized execution records, checks the result-record digest and rich-artifact hash-list digest, regenerates publication tables and figures, and compares them with the bound summaries. The three deterministic repeat seeds must produce identical traces. This check establishes computational reproduction of the controlled evidence; it does not increase its statistical authority.

The exact-contract archive binds all 400 case identities, the protected bundle, canonical decision rows, bootstrap seed, arm counts, and independent verification. The TREC-COVID archive binds the 50 paired topics, route mapping, arm outputs, bootstrap result, and registered gate. These records preserve the failed recall and cost criteria together with the favorable secondary nDCG result.

The MetaSyn and AutoResearchBench stress tests are distributed as bounded result records and evaluator receipts. Public OpenAIRE/Crossref captures are retained with their validity failures and cannot be reinterpreted as negative or null comparisons. Third-party source data remain available from their original providers under those providers' terms.

During double-anonymized review, the blinded supplementary archive is the canonical access route. Author identification and any later public deposit are kept separate from the reviewer manuscript; changing the archived bytes would require a new manifest and affected-claim audit.

9 Ethics

This study collected no new human-participant records or personally identifiable information. The controlled index and exact-contract cases are synthetic. External experiments use publicly released information-retrieval and systematic- review benchmarks. The study evaluates search-control behavior and does not present a deployed clinical, surveillance, or human-facing decision system.

10 Conclusion

Open-world scientific-literature discovery needs separate control of evidence acquisition and closure authority. The method developed here derives route identity from provenance, distinguishes content identity from question-conditioned processing, keeps route stopping local, and retains unresolved material routes as open obligations. Controlled and exact-contract tests show that these rules behave as designed. The registered TREC-COVID recall-and-cost gate nevertheless fails, and favorable nDCG does not replace that decision. The present contribution is therefore a bounded, reproducible control method for representing acquisition authority. External retrieval benefit and open-world completeness remain unestablished.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI ChatGPT to support literature searching, source checking, reproducibility review, manuscript editing, and submission-package preparation. The tool was not treated as an author and did not replace scientific judgment. Result-bearing statements and citations were checked against the archived evidence and primary-source records. The authors remain responsible for the article's content.

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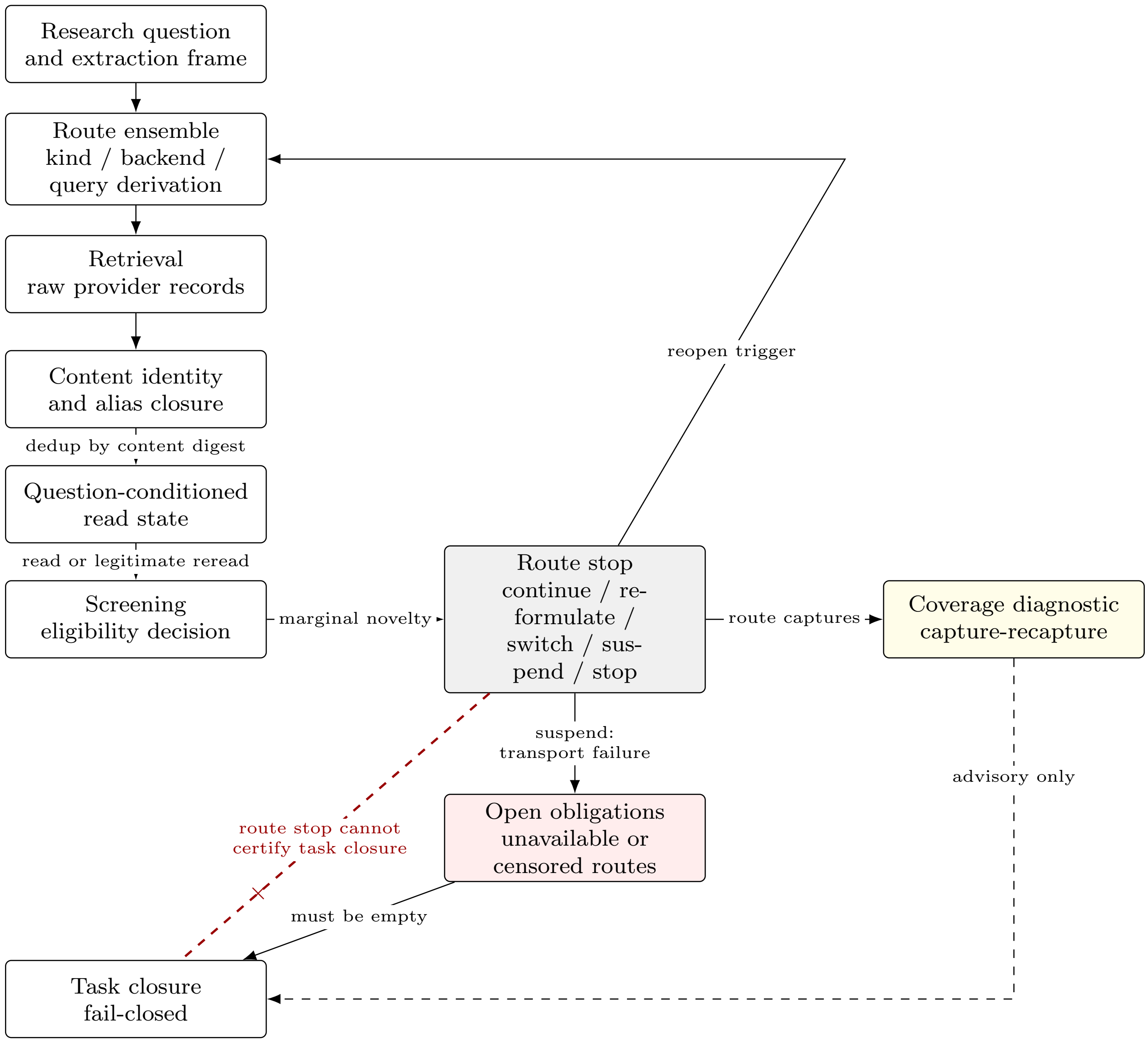

Figure 1: The discovery pipeline separates retrieval, screening, route stopping, and task closure. Solid arrows carry state updates. The dashed coverage diagnostic informs search but does not certify closure, and the crossed arrow is the forbidden promotion from a route stop to task completeness. Transport failure leaves an open obligation rather than evidence of absence.

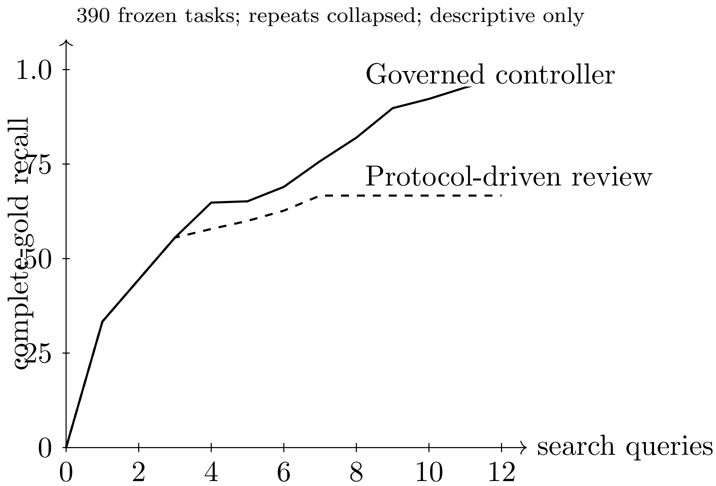

Figure 2: Cumulative complete-gold recall against host-recorded search queries. Deterministic repeats were checked for identical traces and collapsed within task. Curves are task means and descriptive only. The synthetic experiment does not support a wall-clock or monetary-cost claim.

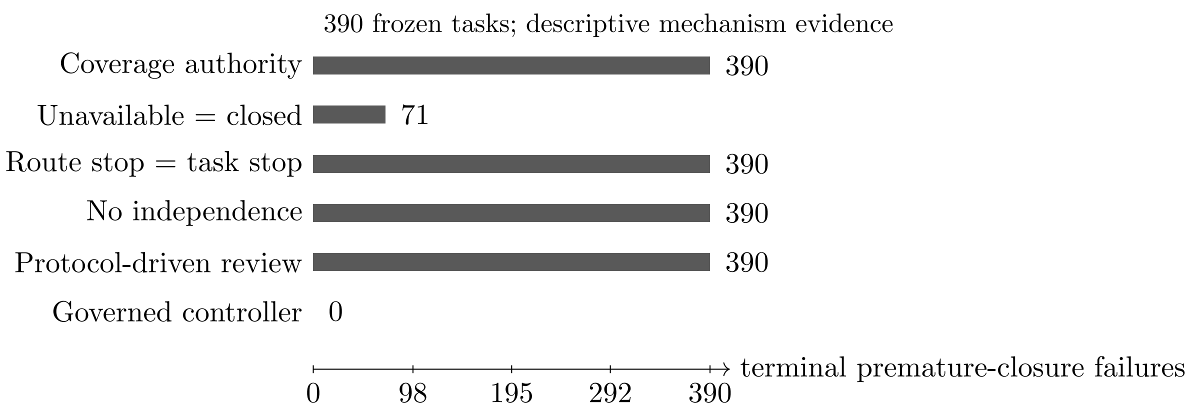

Figure 3: Terminal premature-closure classifications for selected control ablations. Bars are task counts from the frozen complete-gold index, not inferential estimates. The evaluator uses a fixed failure precedence, so a displayed terminal class can mask lower-precedence failures.